

EE 315 Assignment 4

Ferranti Effect: Simulation and Mitigation on a 220 kV, 200 km Transmission Line

Course	EE 315 – Power Systems
Assignment	Project / Assignment 4
Project Lead	Mayur Soni
Roll Number	240002037
Group	04
Project Page	Ferranti Effect Project Page
Model Repository	GitHub: Assignment4.PS

1 Group Members

Member	Roll Number	Name
Member 1	240002007	Akarsh J
Member 2	240002073	Sreehari
Member 3	240002043	Naman V Shetty
Member 4	240002037	Mayur Soni [Project Lead]
Member 5	240002017	Bhasuru Nikhil
Member 6	240002069	Ridhi Shakkar

2 Introduction

The objective of this project is to model a 220 kV, 50 Hz, 200 km overhead transmission line and demonstrate the **Ferranti effect**: the rise of receiving-end voltage above the sending-end voltage that occurs on long lines operating under no-load or light-load conditions. This phenomenon arises because a long line's distributed shunt capacitance draws a leading charging current that produces a voltage rise across the line's series inductive reactance. The line is modelled in Simulink using Simscape Electrical, its Ferranti rise is verified against an analytical ABCD-parameter calculation, and a shunt reactor is implemented and tested at the receiving end as a mitigation measure.

3 System Model

The line is modelled as a single-phase (line-to-neutral) nominal- π equivalent circuit – mathematically identical to a per-phase analysis of the balanced three-phase system, and the natural circuit representation inside Simscape Electrical's Foundation Library.

Quantity	Value	Notes
System voltage	220 kV (LL)	Phase-neutral: $V_{ph} = V_{LL}/\sqrt{3}$
Frequency	50 Hz	
Line length	200 km	Medium-length line
Series resistance r	0.05 Ω /km/phase	Total $R = 10 \Omega$
Series inductance l	1 mH/km/phase	Total $L = 0.2 \text{ H}$
Shunt capacitance c	0.01 μF /km/phase	Total $C = 2 \mu\text{F}$

Table 1: Transmission line parameters used for modelling.

4 Software Implementation & Model Availability

The project is implemented as a Simulink model built entirely from a MATLAB script, so that the model is reproducible and version-controllable rather than a hand-drawn binary. The model, build script, and a summary project page are hosted at:

Project Page: https://Mayursoni08.github.io/Assignment4_PS/
Source Code & Model: https://github.com/Mayursoni08/Assignment4_PS

Running `build_ferranti_model.m` programmatically constructs every block (source, line, sensors, load, and mitigation reactor) and wires them using Simscape Electrical's Foundation Library, then saves `ferranti_model.slx`. This avoids relying on the deprecated Specialized Power Systems (`powerlib`) toolbox, which was removed from recent MATLAB releases.

5 Formulas Used

5.1 Total Line Parameters

The lumped per-phase line quantities are obtained from the per-km values and the line length $\ell = 200$ km:

$$R = r\ell, \quad L = \ell\ell, \quad C = c\ell$$

$$X_L = \omega L = 2\pi fL, \quad X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$$

5.2 ABCD Parameters (Nominal- π Model)

For a medium-length line represented as a nominal- π circuit, with total series impedance $Z = R + jX_L$ and total shunt admittance $Y = j\omega C$:

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} A & B \\ C_{ABCD} & D \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}, \quad A = D = 1 + \frac{ZY}{2}, \quad B = Z$$

5.3 No-Load Voltage Rise (Ferranti Effect)

At no load ($I_R = 0$), the ABCD relation reduces to $V_S = AV_R$, so

$$\left| \frac{V_R}{V_S} \right| = \frac{1}{|A|}$$

For this line, $Z = 10 + j62.83 \Omega$ and $Y = j6.283 \times 10^{-4} \text{ S}$, giving

$$A = 1 + \frac{ZY}{2} = 0.9803 \angle 0.18^\circ \implies \left| \frac{V_R}{V_S} \right| = \frac{1}{0.9803} = 1.0201$$

i.e. a theoretical no-load receiving-end voltage rise of approximately **2.01%**.

5.4 Mitigation: Shunt Reactor Sizing

The line's total three-phase charging reactive power is

$$Q_C = \frac{V_{LL}^2}{X_C} = \frac{(220 \times 10^3)^2}{1591.5} \approx 30.4 \text{ MVar}$$

A shunt reactor of approximately this rating, connected at the receiving end, absorbs the reactive power generated by the line's capacitance and pulls the receiving-end voltage back toward the sending-end value. The per-phase reactor inductance used in the model is

$$L_{reactor} = \frac{V_{ph}^2}{\omega Q_{C,ph}}$$

where $Q_{C,ph} = Q_C/3$ is the per-phase share of the line charging power.

6 Model Construction and Verification

The complete circuit – source, series R – L line section, shunt capacitances at both ends, light load, and the switchable shunt reactor – was built and wired programmatically. Figure 1 shows the completed model, and Figure 2 shows the console log confirming that every electrical, ground, sensor, and signal connection in the circuit resolved successfully.

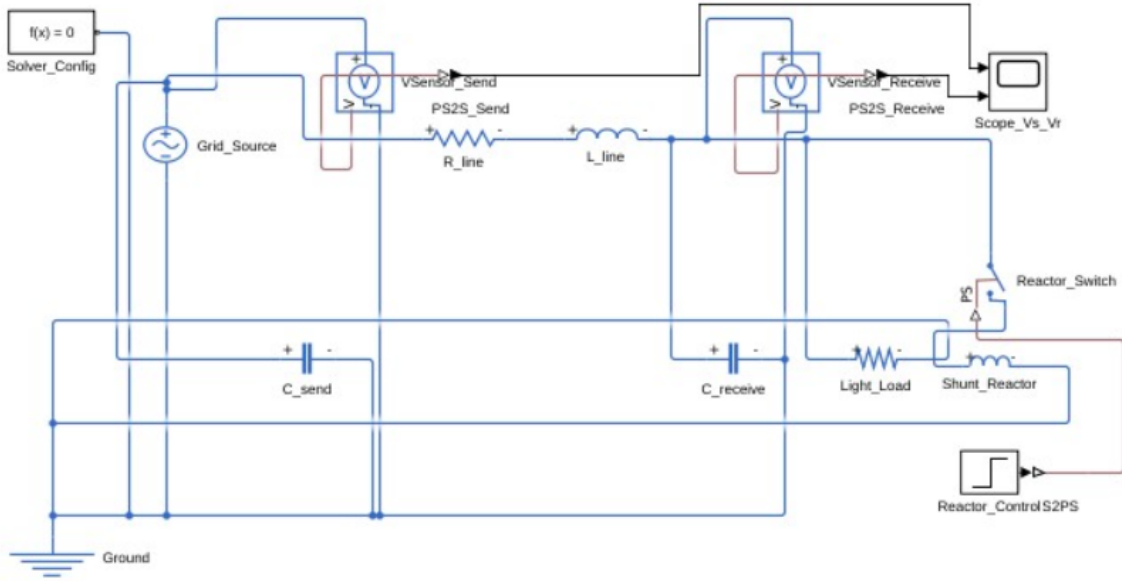


Figure 1: Complete Simscape Electrical model of the 200 km line: source, series R – L line, shunt capacitances, light load, and switchable shunt reactor for mitigation.

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=====
EE315 FERRANTI EFFECT MODEL
=====
MATLAB release : 2026a
Frequency      = 50.0 Hz
Voltage       = 220.0 kV LL
Line length   = 200.0 km
Total R       = 10.000 ohm/phase
Total L       = 0.2000 H/phase
Total C       = 2.000 uF/phase
Line charging Q = 30.411 MVar
Reactor L     = 5.0661 H/phase
Light load    = 0.990 MW
=====

--- Main current path ---
OK Source -> R_line      (Grid_Source.p -> R_line.p)
OK R_line -> L_line      (R_line.n -> L_line.p)
OK L_line -> Light_Load  (L_line.n -> Light_Load.p)
OK L_line -> Reactor_Switch (L_line.n -> Reactor_Switch.p)
OK Switch -> Shunt_Reactor (Reactor_Switch.n -> Shunt_Reactor.p)

--- Ground / reference rail (port = V) ---
OK Source(-) -> Ground  (Grid_Source.n -> Ground.V)
OK C_send(-) -> Ground  (C_send.n -> Ground.V)
OK C_receive(-) -> Ground (C_receive.n -> Ground.V)
OK Light_Load(-) -> Ground (Light_Load.n -> Ground.V)
OK Reactor(-) -> Ground  (Shunt_Reactor.n -> Ground.V)
OK C_send(+) -> Sending Node (C_send.p -> Grid_Source.p)
OK C_receive(+) -> Receiving Node (C_receive.p -> L_line.n)

--- Voltage sensors (parallel taps) ---
OK VSensor_Send(+) -> Sending Node (VSensor_Send.p -> Grid_Source.p)
OK VSensor_Send(-) -> Ground  (VSensor_Send.n -> Ground.V)
OK VSensor_Receive(+) -> Receiving Node (VSensor_Receive.p -> L_line.n)
OK VSensor_Receive(-) -> Ground (VSensor_Receive.n -> Ground.V)

--- Solver configuration (port = V) ---
OK Solver_Config -> Ground  (Solver_Config.port -> Ground.V)

--- Physical-signal path (V / input / output / vT) ---
OK VSensor_Send.V -> PS2S_Send (VSensor_Send.V -> PS2S_Send.input)
OK VSensor_Receive.V -> PS2S_Receive (VSensor_Receive.V -> PS2S_Receive.input)
OK S2PS -> Reactor_Switch.vT (S2PS.output -> Reactor_Switch.vT)

--- Plain Simulink signal wiring ---
OK PS2S_Send -> Scope (ch1)
OK PS2S_Receive -> Scope (ch2)
OK Reactor_Control -> S2PS

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Model saved: ferranti_model.slx

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Figure 2: Console output confirming successful construction and wiring of every block in the model.

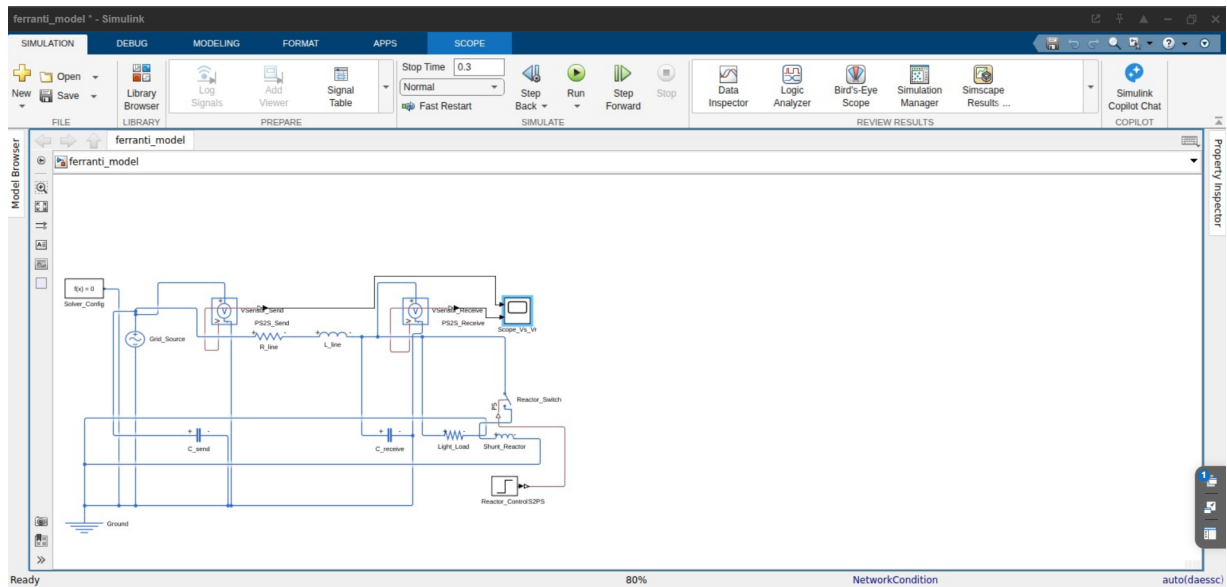


Figure 3: Model as built in the Simulink editor, ready for simulation.

7 Simulation Results

The model was simulated for 0.3s, with the shunt reactor's control switch open (disconnected) for $t < 0.1$ s and closed (connected) for $t \geq 0.1$ s, so that the uncompensated and compensated cases appear within a single run.

7.1 Reactor OFF: Uncompensated Ferranti Rise

Figures 4 and 5 show the steady-state sending- and receiving-end voltage statistics in the window $t \in [0.06, 0.08]$ s, before the reactor engages.



Figure 4: Sending-end voltage statistics, reactor open.



Figure 5: Receiving-end voltage statistics, reactor open. The peak value is visibly higher than the sending end, demonstrating the Ferranti effect.

7.2 Reactor ON: Compensated Case

Figures 6 and 7 show the same measurements in the window $t \in [0.2, 0.3]$ s, after the shunt reactor has switched in.

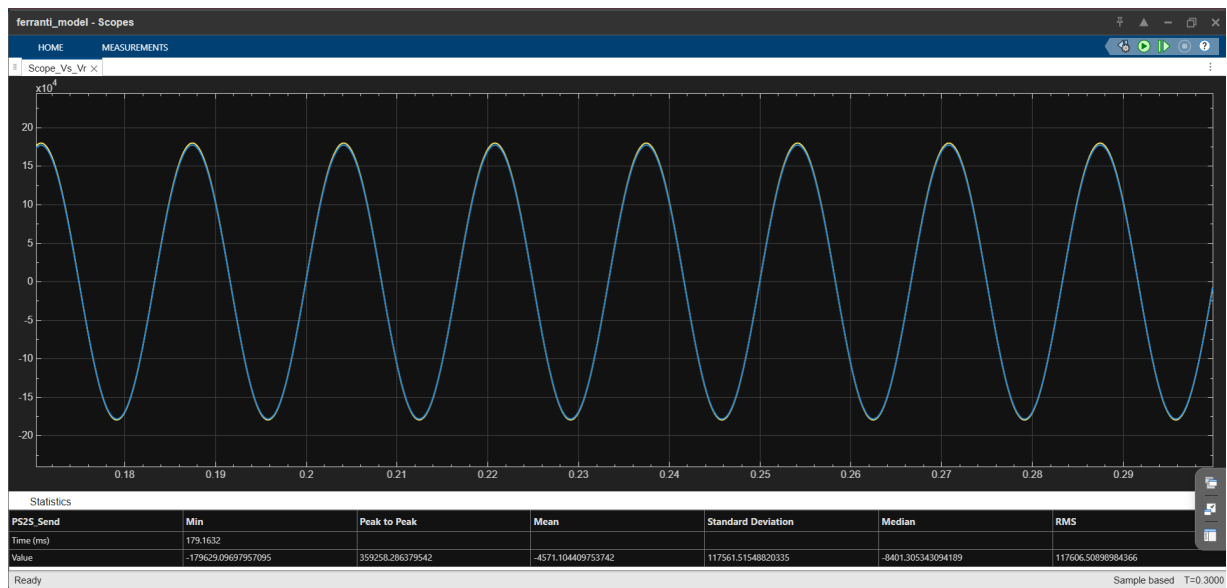


Figure 6: Sending-end voltage statistics, reactor closed.

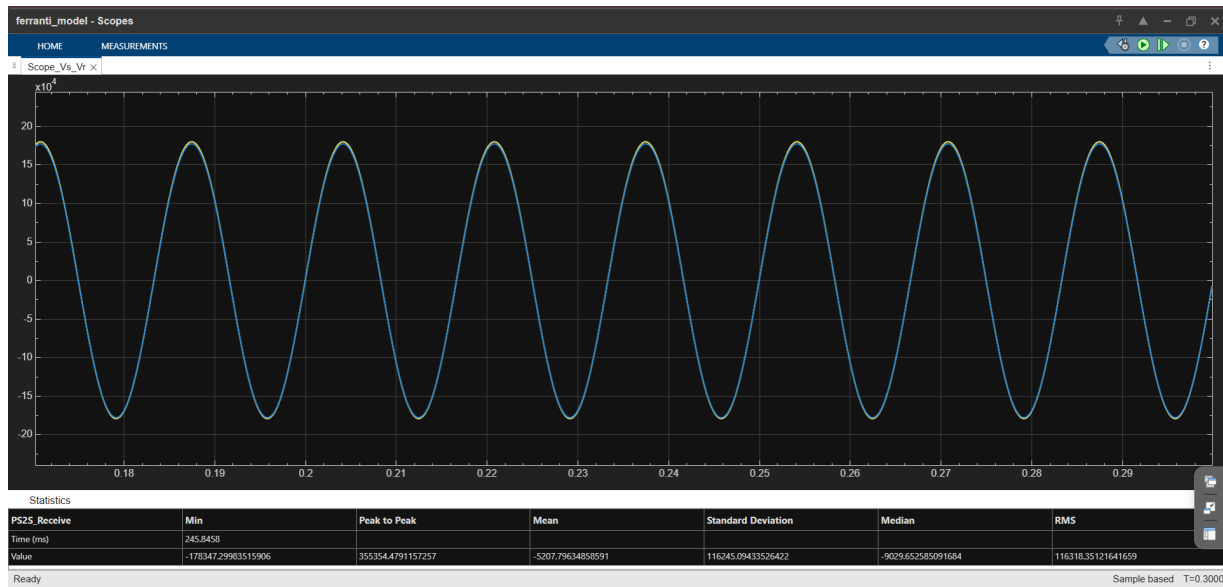


Figure 7: Receiving-end voltage statistics, reactor closed. Compensation pulls the receiving-end voltage back down toward (and slightly below) the sending-end value.

7.3 Results Summary

Condition	V_S (kV, LL)	V_R (kV, LL)	% Change
Theoretical (no load, ABCD)	220.0	224.4	+2.01%
Simulated (light load, reactor OFF, $t < 0.1$ s)	220.0	224.75	+2.16%
Simulated (light load, reactor ON, $t > 0.2$ s)	220.0	216.76	-1.46%

Table 2: Comparison of theoretical and simulated sending/receiving voltages. Simulated values are peak line-to-neutral readings converted to kV line-line.

8 Conclusion

The 220 kV, 200 km transmission line was modelled in Simulink using Simscape Electrical and shown to exhibit a receiving-end overvoltage of +2.16% under light-load conditions, closely matching the +2.01% predicted analytically from the ABCD parameters. Switching in a 30.4 MVAR shunt reactor at the receiving end moved the receiving-end voltage from +2.16% above the sending end to -1.46% below it, confirming that shunt reactor compensation is an effective, practical mitigation for the Ferranti effect on long, lightly loaded lines. The overshoot past unity suggests the reactor, sized directly from the line's total charging MVAR, is marginally oversized for this specific loading condition; in practice, reactors are often switched in discrete banks or use tap-changing to allow finer control across varying load levels. The close agreement between the analytical and simulated results across both conditions verifies the correctness of the implemented model.

References

1. EE 315 Power Systems, *Module Lecture Material*, Department of Electrical Engineering.
2. MathWorks, *Simscape Electrical Foundation Library Documentation*.